Metabolic Response of Different High-Intensity Aerobic Interval Exercise Protocols

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Abstract

Gosselin, LE, Kozlowski, KF, DeVinney-Boymel, L, and Hambridge, C. Metabolic response of different high-intensity aerobic interval exercise protocols. *J Strength Cond Res* 26(10): 2866–2871, 2012—Although high-intensity sprint interval training (SIT) employing the Wingate protocol results in significant physiological adaptations, it is conducted at supramaximal intensity and is potentially unsafe for sedentary middle-aged adults. We therefore evaluated the metabolic and cardiovascular response in healthy young individuals performing 4 high-intensity (~90% $\dot{V}O_2$max) aerobic interval training (HIT) protocols with similar total work output but different work-to-rest ratio. Eight young physically active subjects participated in 5 different bouts of exercise over a 3-week period. Protocol 1 consisted of 20-minute continuous exercise at approximately 70% of $\dot{V}O_2$max, whereas protocols 2–5 were interval based with a work-active rest duration (in seconds) of 30/30, 60/30, 90/30, and 60/60, respectively. Each interval protocol resulted in approximately 10 minutes of exercise at a workload corresponding to approximately 90% $\dot{V}O_2$max, but differed in the total rest duration. The 90/30 HIT protocol resulted in the highest $\dot{V}O_2$, HR, rating of perceived exertion, and blood lactate, whereas the 30/30 protocol resulted in the lowest of these parameters. The total caloric energy expenditure was lowest in the 90/30 and 60/30 protocols (~150 kcal), whereas the other 3 protocols did not differ (~195 kcal) from one another. The immediate postexercise blood pressure response was similar across all the protocols. These finding indicate that HIT performed at approximately 90% of $\dot{V}O_2$max is no more physiologically taxing than is steady-state exercise conducted at 70% $\dot{V}O_2$max, but the response during HIT is influenced by the work-to-rest ratio. This interval protocol may be used as an alternative approach to steady-state exercise training but with less time commitment.

Key Words lactate, heart rate, blood pressure, training, RPE

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Introduction

In 2008, the U.S. Department of Health and Human Services recommended that all Americans perform at least 150 minutes of moderate-intensity (3–6 METS) physical activity each week (8) to improve health. If performed over 5 days, individuals would need to exercise for 30 minutes daily. However, lack of time is cited as a common reason by most Americans as to why they do not exercise regularly (4).

Over the last few years, a number of high-intensity sprint interval training (SIT) studies using the Wingate protocol have shown that significant physiological benefits can be obtained by performing as little as 6–8 minutes of exercise per week (5,7) and that these changes are comparable with those found using traditional aerobic conditioning for 200 min·wk^{-1} (6). Moreover, the use of SIT has resulted in significant improvements in glucose regulation and insulin sensitivity (1) and flow-mediated dilation (13).

Although the use of SIT induces significant physiological and functional improvements from only 2 to 3 minutes of exercise per workout, the total time commitment remains at 20–30 minutes excluding warm-up and cool-down because of the prolonged rest period between each work interval. Thus, in terms of time efficiency, SIT is no more effective than traditional aerobic training is. Perhaps more important, SIT employing the Wingate protocol is conducted at supramaximal intensity (load equal to 7.5% of subject body weight) and is potentially unsafe for sedentary middle-aged and senescent subjects.

Aerobic interval training protocols using exercise intensities ≤100% $\dot{V}O_2$max have been shown to significantly improve aerobic capacity and selected markers of health in both normal individuals and in diseased patients (11,15,17) and in some cases have been shown to be superior to traditional aerobic training (17). However, the optimal prescription with regards to intensity, work and rest duration, and number of intervals have yet to be elucidated. If aerobic interval training is to be optimally prescribed to individuals, it is necessary to understand the metabolic, cardiovascular, and perceptual responses of different protocols conducted at the same external workload but with varying work-to-rest durations. The purpose of this study, therefore, was to evaluate the metabolic (oxygen consumption, caloric energy expenditure, and blood lactic acid [LA]) and cardiorespiratory (heart rate [HR] and blood pressure) responses of healthy young individuals to 4 different aerobic interval training protocols performed at 90% $\dot{V}O_2$max and to compare them with a traditional exercise protocol at 70% $\dot{V}O_2$max. The interval-based protocols resulted in a similar total work output but differed in the work-to-rest ratio. We hypothesized that increasing the work-to-rest ratio, while conducting similar amounts of work, would increase the metabolic, cardiovascular, and perceptual response to interval exercise. Our ultimate goal is to identify a safe high-intensity aerobic interval-based exercise program that reduces the time of exercise required for physiological and functional improvement yet remains safe for sedentary middle-aged individuals.

Methods

Experimental Approach to the Problem

To test our hypothesis, a repeated measures design was used. After the determination of the subject's $\dot{V}O_2$max, the subjects participated in 5 different exercise protocols over a 3-week period with at least 3 days of rest between each exercise session, including the $\dot{V}O_2$max test. All testing was performed in the late afternoon. For a given subject, each exercise protocol was performed within 1 hour of the same time of the day. The subjects were asked to refrain from vigorous exercise for 24 hours before each test and to maintain their normal diet throughout the study. Protocol test order was randomly assigned for each subject. The independent variable was exercise protocol, whereas the major dependent variables included oxygen consumption ($\dot{V}O_2$), HR, blood lactate, and rating of perceived exertion (RPE).

Subjects

Eight healthy physically active subjects (2 men, 6 women) aged 20–30 years participated in the study. The subjects averaged 23.1 ($\pm$2.1) years of age and 64.3 ($\pm$6.6) kg. In addition, the mean $\dot{V}O_2$max was 48.5 ($\pm$6.0) ml·kg^{-1}·min^{-1}. The University at Buffalo Institutional Review Board approved all the procedures. All the subjects provided informed consent before participation in the study.

Procedures

Each subject first performed a running graded exercise test on a motor driven treadmill for determination of $\dot{V}O_2$max. The workload was increased every 2 minutes by increasing either the speed or grade. The test was terminated when the subjects reached volitional exhaustion. Verbal encouragement was provided throughout the test. Oxygen consumption was continually monitored breath by breath through a plastic face mask connected to a portable metabolic cart (Oxycon Mobile, VIASYS; Yorba Linda, CA, USA) via a low dead space (30-ml) bidirectional digital volume transducer (Triple V Assembly, Jaeger). A mesh headgear was fit over the crown of the subject's head and connected at 4 points to the facemask and turbine assembly. The participants were fit with a lightweight nylon vest harness holding the telemetric units. The total weight of all components was approximately 950 g. The participants also wore a chest strap (Kempele, Finland) to monitor HR. Subjective RPE was recorded at the end of each workload using the standard 6–20 Borg scale (3).

Each subject participated in 5 different exercise protocols over a 3-week period with at least 3 days of rest between each exercise session. Protocol test order was randomly assigned for each subject. Protocol 1 consisted of the subject exercising continuously at a workload corresponding to approximately 70% of $\dot{V}O_2$max for 20 minutes. Protocols 2–5 were interval based with work-active rest duration (in seconds) of 30/30, 60/30, 90/30, and 60/60, respectively. For protocols 2–5, the subjects exercised at a workload corresponding to 90% $\dot{V}O_2$max, as determined from the graded exercise test. Each work bout was followed by an active rest period whereby subjects walked on the treadmill at a workload equal to approximately 35–40% $\dot{V}O_2$max. Each interval protocol resulted in approximately 10 minutes of work but differed slightly in total duration because of the varied rest duration. Oxygen consumption was continuously monitored during each exercise protocol as previously described. The HR and RPE were also monitored during each exercise protocol.

At the end of the last work interval of each protocol, the treadmill speed was reduced to 2 mph at a 0% grade for 3 minutes. Blood pressure was taken manually within 30 seconds after exercise completion. In addition, 30–60 seconds after each protocol, approximately 50–100 μl of blood was collected via a finger stick for the determination of blood LA using the Accutrend® Lactate Analyzer (Roche Diagnostics, Mannheim, Germany). Before collection, the finger was cleansed with alcohol and allowed to air dry. Total caloric energy expenditure during each protocol was determined by multiplying the mean $\dot{V}O_2$ by the caloric equivalent for the mean respiratory exchange ratio.

Statistical Analyses

All data were analyzed using repeated measures analysis of variance (Sigma Stat, version 3.0). A p value of $\leq$0.05 was considered statistically significant. When significant differences existed, pairwise multiple comparisons procedure

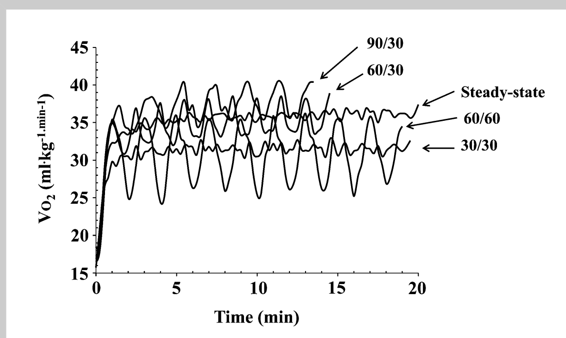

Figure 1. Mean oxygen consumption as a function of time during each exercise protocol.

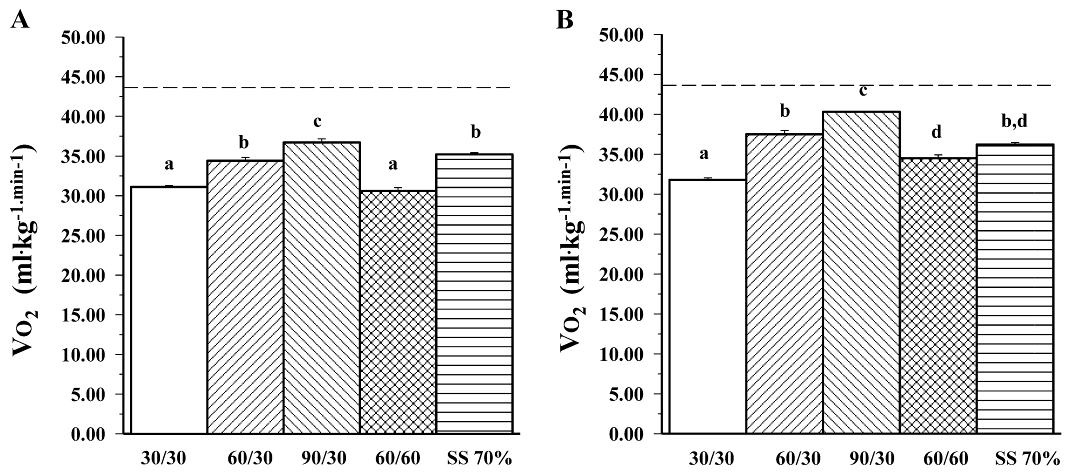

Figure 2. Mean (±*SD*) (A) and peak (B) oxygen consumption of each exercise protocol. Bars with similar letters are not significantly different from one another. The dotted line represents the $\dot{V}O_2$ at 90% of maximal aerobic capacity.

(Holm-Sidak) method) was used to determine which protocols differed from one another. The power for the primary outcome measure of peak $\dot{V}O_2$ with a sample size of 8 and a $p \leq 0.05$ was >0.9.

Results

The mean oxygen consumption ($\dot{V}O_2$) as a function of time for each exercise protocol, including rest and work periods, is illustrated in Figure 1. The mean $\dot{V}O_2$ for each protocol, encompassing both work and rest periods, is illustrated in Figure 2A. The mean $\dot{V}O_2$ was significantly higher in the 90/30 protocol. In addition, the mean $\dot{V}O_2$ for the 60/30 and steady-state protocols did not differ from one another but was significantly higher than the 30/30 or 60/60 protocol. Lastly, the mean $\dot{V}O_2$ was lowest in the 30/30 and 60/60 protocols, and these 2 protocols did not differ from one another.

For the interval protocols, the peak $\dot{V}O_2$ was determined by averaging the highest $\dot{V}O_2$ at the end of the last 3 work protocols, whereas for the steady-state protocol, the peak $\dot{V}O_2$ represents the mean $\dot{V}O_2$ of the last 3 minutes of exercise. As seen in Figure 2B, the peak $\dot{V}O_2$ was significantly higher in the 90/30 protocol, whereas the 30/30 protocol elicited the lowest value. The peak $\dot{V}O_2$ in the 60/30 protocol was

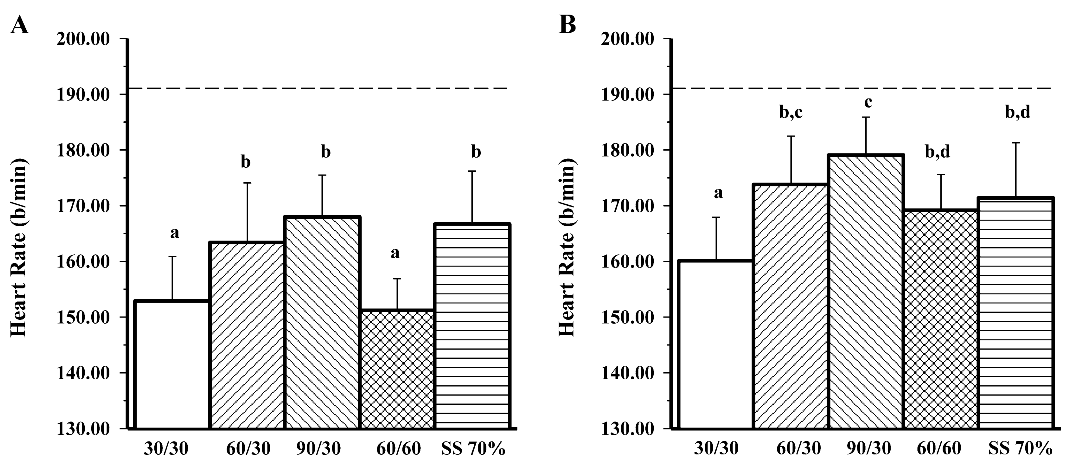

Figure 3. Mean (±*SD*) (A) and peak (B) heart rate (HR) measured at rest and immediately at the end of each exercise protocol. Bars with similar letters are not significantly different from one another. The dotted line represents the mean HR at $\dot{V}O_2$max.

significantly lower than in the 90/30 protocol but significantly higher than the 60/60 or steady-state protocol. In all interval protocols, the peak $\dot{V}O_2$ fell below the targeted $\dot{V}O_2$ of 90% $\dot{V}O_2$max.

The mean HR for each protocol, encompassing both work and rest periods, is illustrated in Figure 3A. The mean HR was significantly lowest in the 30/30 and 60/60 protocols, and these 2 protocols did not differ from one another. The mean HR response was similar in the 60/30, 90/30 and steady-state protocols. For the interval protocols, the peak HR was determined by averaging the highest HR at the end of the last 3 work protocols, whereas for the steady-state protocol, the peak HR represents the mean HR of the last 3 minutes of exercise (Figure 3B). The peak HR was the highest in the 90/30 protocol and lowest in the 30/30 protocol. The peak HR did not differ between the 60/30 and 90/30 protocols. Compared with the 90/30 protocol, the peak HR was significantly lower in the 60/60 and steady-state protocols, whereas these latter 2 protocols did not differ from the 60/30 protocol.

The mean resting blood pressure was 116.5/68 mm Hg (Table 1). At the end of 20 minutes steady-state exercise performed at 70% $\dot{V}O_2$max, the mean blood pressure was 148/68.7 mm Hg. Blood pressure at the end of the interval protocols conducted at 90% $\dot{V}O_2$max did not significantly differ from that of steady-state exercise.

The mean blood lactate concentration, taken at rest and at the end of each exercise protocol, is shown in Figure 4. Blood lactate concentration was significantly higher after all exercise protocols compared with that for rest. There was no significant difference in blood lactate between the 30/30, 60/60, and steady-state protocols. In addition, there was no significant difference in blood lactate concentration between the 60/30 and 90/30 protocols. However, the 90/30 protocol elicited a significantly higher blood lactate than did the 30/30, 60/60, and steady-state protocols. Additionally, blood lactate was significantly higher in the 60/30 protocol compared with that in the 30/30 and steady-state protocols.

Mean total caloric energy expenditure for the 30/30, 60/60, and steady-state protocols averaged approximately 195 kcal, and the total caloric energy expenditure in these 3 protocols was significantly higher than in the 60/30 or 90/30 protocol (~150 kcal). Lastly, there was no difference in energy expenditure between the 60/30 and 90/30 protocols.

For all exercise protocols, the RPE gradually increased with time. The mean peak RPE taken at the end of the exercise

Table 1. Mean ($\pm SD$) systolic and diastolic blood pressures at rest and at the end of each exercise protocol.

Condition	Systolic blood pressure (mm Hg)	Diastolic blood pressure (mm Hg)
Rest	116.5 ± 5.4*	68.0 ± 9.3
30/30	146.3 ± 5.9	70.0 ± 6.0
60/30	150.3 ± 15.9	71.0 ± 6.4
90/30	152.5 ± 11.7	69.0 ± 5.0
60/60	153.1 ± 10.5	72.7 ± 6.0
Steady state	148.0 ± 9.8	68.7 ± 5.8

*Significantly different from the other conditions, $p < 0.05$.

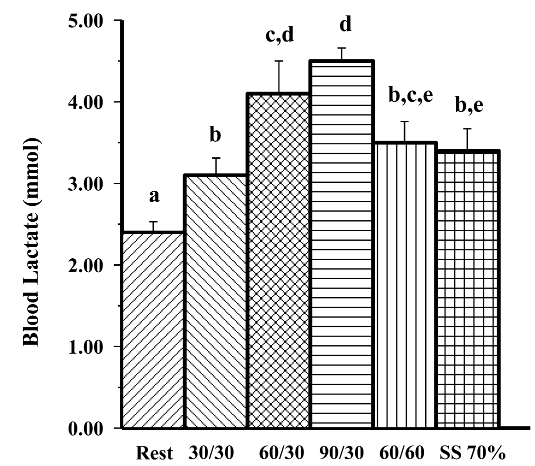

Figure 4. Mean ($\pm SD$) blood lactate measured at rest and immediately at the end of each exercise protocol. Bars with similar letters are not significantly different from one another.

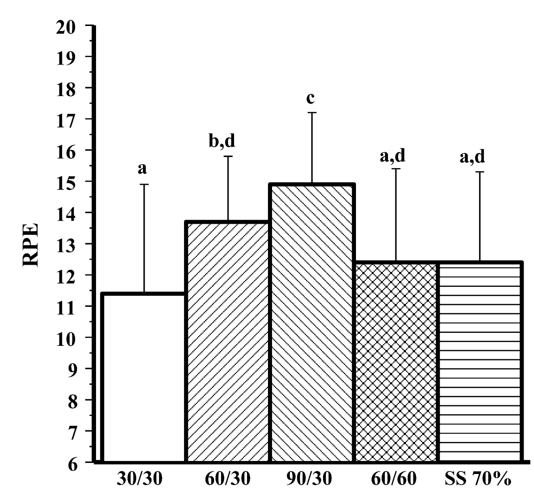

Figure 5. Mean ($\pm SD$) rating of perceived exertion (RPE) measured at the end of each exercise protocol. Bars with similar letters are not significantly different from one another.

protocol appears in Figure 5. The final RPE was significantly higher in the 90/30 protocol compared with all others. The final RPE did not differ between the 30/30, 60/60, and steady-state protocols. The final RPE in the 60/30 protocol was significantly greater than the 30/30 but not significantly different from the 60/60 or steady-state protocol.

Discussion

Our present study demonstrates that altering the work-to-rest ratio significantly alters the metabolic and HR response to interval exercise, despite the subjects performing similar amounts of work. Our findings highlight that when using this exercise paradigm, the subjects can achieve several minutes of high-intensity exercise without undue discomfort or fatigue.

The mean and peak $\dot{V}O_2$ achieved during the 30/30 protocol averaged 64.1 and 65.6% $\dot{V}O_2$max, respectively, whereas during the 60/30 protocol, the mean and peak $\dot{V}O_2$ averaged 70.9 and 77.3% $\dot{V}O_2$max, respectively. During the 90/30 protocol, the mean and peak $\dot{V}O_2$ averaged 75.7 and 83.1% $\dot{V}O_2$max, respectively. Thus, the percent of $\dot{V}O_2$max reached during interval exercise performed at a workload corresponding to 90% $\dot{V}O_2$max was influenced by the work-to-rest ratio, that is, the greater the work-to-rest ratio, the greater the $\dot{V}O_2$. Interestingly, however, the $\dot{V}O_2$ (mean or peak) never reached the value observed during the continuous graded exercise test in any of the interval protocols.

At the beginning of exercise, energy demand is met through a number of different pathways, including the immediate breakdown of high energy phosphate stores (e.g., adenosine triphosphate, creatine phosphate), glycolysis, and aerobic metabolism (12), and the relative contribution of these pathways depends on a number of variables. The contribution of aerobic metabolism is reflected by oxygen consumption during work, and the extent of rise is dependent not only upon the duration of exercise or the level of fitness within an individual (10) but also on the work intensity. Whipp and Wasserman (16) demonstrated that at low workloads, steady-state $\dot{V}O_2$ could be reached within 3 minutes from the start of exercise. With high work intensities, however, the $\dot{V}O_2$ kinetics was characterized by an initial rapid phase followed by a slower secondary phase such that steady-state $\dot{V}O_2$ was not achieved even in the sixth minute of exercise. Given that our subjects exercised at a workload corresponding to 90% $\dot{V}O_2$max with regular intermittent rest periods, it seems likely that the work-to-rest dynamics of the exercise protocol did not allow our subjects sufficient time to achieve the expected $\dot{V}O_2$. At present, it is unknown if increasing the number of work intervals will result in a higher mean or peak $\dot{V}O_2$, or if the $\dot{V}O_2$ response will vary significantly in untrained individuals as compared with trained individuals.

The HR response in most respects mirrored the $\dot{V}O_2$ response during the exercise protocols. Perhaps most striking was that the mean HR was significantly lower in the 30/30 and 60/60 protocols as compared with 20 minutes of steady-state exercise conducted at 70% $\dot{V}O_2$max. Heart rate averaged approximately 153 b·min^{-1} during both the 30/30 and 60/60 protocols, and approximately 167 b·min^{-1} during the steady-state protocol. This occurred even though the total caloric energy expenditure was similar across the 3 protocols. By the end of the steady-state protocol, the HR(peak) averaged 171 b·min^{-1}. Additionally, the mean HR during either the 60/30 or 90/30 protocol was not significantly different than the steady-state protocol. With respect to the peak HR achieved during the interval exercise, only the 90/30 protocol was significantly higher than the steady-state protocol. Moreover, the blood pressure response did not differ across all protocols. These findings indicate that high-intensity interval exercise (at workloads corresponding to 90% $\dot{V}O_2$max) employing a protocol described in this study can likely be carried out as safely as traditional steady-state exercise performed at 70% $\dot{V}O_2$max.

We determined whether the interval exercise protocols used in this study resulted in excessively high concentrations of blood lactate, which could ultimately influence work tolerance. Blood lactate concentration was highest after the 90/30 protocol but only averaged 4.5 mmol. Lowering the work-to-rest ratio from 3:1 (90/30) to 1:1 (60/60 or 30/30) resulted in a significantly lower blood lactate concentration, even though the subjects performed similar amounts of work. Ballor and Volovsek (2) previously noted that increasing the work-to-rest ratio significantly increased $\dot{V}O_2$, HR, and blood lactate when the subjects performed interval work at the same intensity. Our findings indicated a similar but modest relationship between blood lactate and work-to-rest ratios. Compared with the steady-state protocol, only the 60/30 and 90/30 protocols resulted in a higher lactate concentration, but even these concentrations were modest at best.

Perceived effort during exercise is an important (but not sole) determinant in a subject's ability or willingness to complete an exercise session. Consistent with the work by Green et al. (9), the RPE gradually increased with time in all 5 exercise protocols. Of the 4 interval protocols performed in this study, only the 90/30 protocol was rated higher than steady-state exercise conducted at 70% $\dot{V}O_2$max, whereas the RPE during the steady-state protocol did not differ from the 30/30, 60/30, or the 60/60 protocol. Thus, the high interval training (HIT) protocols with a work-to-rest duration of 60/30, 60/60, or 30/30 can be performed by young subjects without undue discomfort. Whether or not older sedentary individuals respond in a similar manner is yet to be determined.

Sprint interval training using the Wingate protocol has been studied extensively during the last several years. Although short-term studies have documented significant improvements in work endurance and peak power output (5,7), oxidative enzyme potential and lipid oxidation (6), and flow-mediated dilation (13), little to no changes in aerobic capacity have been observed (5,7). Moreover, because this exercise is conducted at supramaximal intensity, this type of training is likely not feasible for sedentary older individuals who may be at risk for cardiovascular or metabolic disease. However, other types of high-intensity interval training paradigms have been used for normal and patient populations, which lead to significant improvements in aerobic capacity. For example, Wisloff et al. (17) had heart patients train at approximately 95% of peak HR 3 times per week for 12 weeks using an interval protocol consisting of four 4-minute work intervals interspersed with 12 3-minute rest intervals conducted at 50–70% of peak HR. After interval training, peak $\dot{V}O_2$ increased 46%. Using a similar training paradigm, Slordahl et al. (14) reported that $\dot{V}O_2max$ increased 18% in young healthy subjects. However, in both these studies, the total training duration exceeded 30 minutes per workout and therefore may not be construed as time effective by a majority of Americans.

Recently, Little et al. (11) employed a similar HIT protocol as described in this study in healthy young men. Each subject trained at approximately 100% of peak power output and performed 8- to 12 60-second intervals interspersed with 75-second recovery periods (3 $d \cdot wk^{-1} \times 2$ weeks). The total time commitment, including warm-up and cool-down was approximately 20–29 minutes depending upon the number of intervals performed. The results indicate that cycling endurance, muscle oxidative capacity, resting muscle glycogen and total GLUT 4 content significantly increased. Their findings indicate that a lower volume HIT paradigm is effective in improving physiological function. We propose that the total duration of exercise may be reduced to <20 minutes by employing the 60/30 protocol described in this study. Whether or not this protocol will lead to significant training adaptations in cardiorespiratory endurance or muscle metabolism remains to be established, but recent data by Little et al. (11) are encouraging.

Practical Applications

Because exercise intensity is typically titrated according to the HR response, exercise specialists should consider the work-to-rest ratio, and work and rest duration when prescribing high-intensity aerobic interval exercise. For example, if using the protocol described in this study, a lower work-to-rest ratio could be used for sedentary middle-aged subjects yet still obtain the same amount of mechanical work. As aerobic capacity and exercise tolerance improves, the subjects could be progressed to a higher work-to-rest ratio but with fewer intervals, thereby shortening the total workout duration. Whether or not similar training adaptations will be observed between the different protocols (e.g., 90/30, 60/30, etc.) remains to be determined.

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